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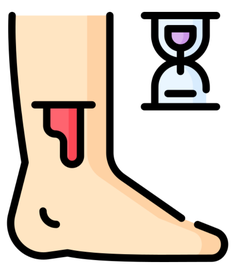
Exploring Correlation Between Tissue Composition and Healing in Chronic Wounds

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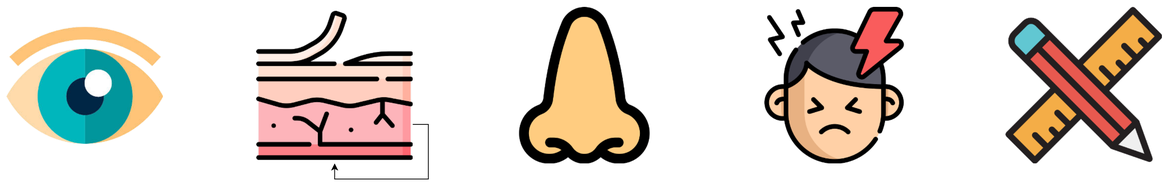
Chronic Wound Healing

Chronic Wounds



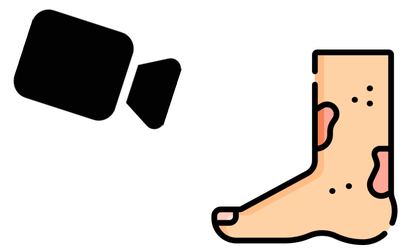
Do not heal in expected time

Current Assessment Parameters



Subjective and difficult to track over time

Objective



Correlate tissue composition with healing for an objective comparison

Data Collection and Labelling

Data Filtering

Data collected from hospital
66 Patients, 500 Images

Filter for relevant patients
35 Patients, 73 Wounds, 373 Images

Data Labelling

Annotate various tissues

Calculate wound composition and area

Analysis

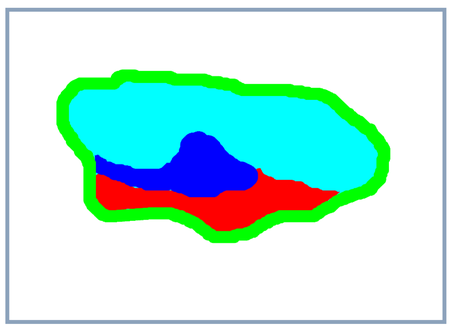
Correlate tissue composition and healing evolution

Feature Description

Wound Image



Wound Tissue Labels



Wound Evolution



Wound Properties

Granulation Healthy Red	16.54
Granulation Unhealthy Anergic	19.32
Hard Necrosis	0
Necrotic Slough	64.13
New Epithelium	0
Approximate Wound Area	35.07mm^2

Next Steps

- Further filter for wounds with decreasing area.
- Among healing wounds, establish direct and indirect correlations for various tissues.
- Explore correlation between type of border and healing.

[1] A. Fadzil and D. Kumar, "Skin Image Analysis Granulation Tissue for Healing Assesment of Chronic Ulcers", *Optical Imaging for Biomedical and Clinical Applications*. CRC Press, 2017, Sec 2.1.2.
[2] Maxim Tkachenko, undefined., et al, "Label Studio: Data labeling software," 2020-2022.
[3] Nejati, H., Ghazijahani, H., Abdollahzadeh, M., Malekzadeh, T., Cheung, N.M., Lee, K.H., & Low, L.L. (2018). Fine-grained wound tissue analysis using deep neural network. In *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*.

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Machine Learning Data Analytics